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We affirm that this project is original and is our own work.

Signatures:

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Introduction: The focus of this project was to analyze ice concentration data for the great lakes mainly during the winter seasons of 2023 to 2024, alongside 2022 to 2023 and 2021 to 2022 using C programming and GLUPlot. Using data recorded by NOAA, we performed calculations to determine important ice concentration statistics and to generate graphs to visually showcase trends in the concentrations, sometimes also drawing conclusions based on the noticeable trends. To effectively work on this project, the questions were divided equally among the group members, with each member working on their chosen part while also working as a group to help each other when questions arose. In this report, a procedure paragraph briefly explaining the code and the process of completing the question is included, along with the output of the code, a screenshot of the output of the code and any required graphs and tables. The report is organized in a clean and easy to read manner, ensuring there are no problems when reading the results of our code.

Question 1:

Procedure:

The primary goal of this question was to compute the average of the ice concentration percentages for each of the six lakes as well as the Great Lakes total for the years 2023 and 2024. The data was stored in a file named data.dat, which contains daily records of ice concentrations with details of year, day and percentage value for each of the lakes. To compute this question we developed a function called (calc_average(int year, float avg[]) in the C program. This function opens the data file and skips 7 header lines using (fgets) loop and extracts the relevant data using sscanf. Then it filters entries for the specified year and sum the concentrations for each lake and count valid days. The averages are then calculated by dividing the sums over the day count and Great Lakes Total computed as the mean of six lake averages. In the main program, the function was called for 2023 and 2024 and the results were displayed in a formatted table.

Output screenshot:

```
Question 1:
Yearly Average Ice Concentration Percentages:
Lake          | 2023 Ave % | 2024 Ave %
-----
Superior      | 0.56       | 2.15
Michigan      | 0.11       | 3.45
Huron         | 0.14       | 6.08
Erie          | 0.00       | 3.33
Ontario       | 0.04       | 0.65
St. Clair     | 0.02       | 15.38
GL Total      | 0.15       | 5.17
```

Question 2:

Procedure:

The main goal of this question was to identify the highest and lowest average Ice concentration for 2023 and 2024 of all six lakes separately. This question was solved directly on the averages calculated in Question 1. The input consists of two arrays (avg_2023) and (avg_2024) which each was storing six lake averages from the previous question. Later in the main program we compared average values by using loops. The process began by initializing the maximum and minimum values of each lake in the specified year, then iterating through the array to update these extremes and their corresponding lake names which were stored in predefined array

listings. The results were as a statement highlighting the highest and lowest average Ice concentrations for each lake in the specified year.

Output:

The Highest and Lowest Ice Concentrations:

2023 - Highest: Superior (0.56%), Lowest: Erie (0.00%)

2024 - Highest: St. Clair (15.38%), Lowest: Ontario (0.65%)

	Highest	Lowest
2023	Superior - 0.56%	Erie - 0.00%
2024	St. Clair - 5.38%	Ontario - 0.65%

Output screenshot:

```
Question 2:

The Highest and Lowest Ice Concentrations:
2023 - Highest: Superior (0.56%), Lowest: Erie (0.00%)
2024 - Highest: St. Clair (15.38%), Lowest: Ontario (0.65%)
```

Question 3:

Procedure:

For this question, we had to find the days where the highest ice concentration was recorded for each lake. The program displays the highest concentration and the date for this recorded value. The program reads the ice concentrations from the data file using `read_data()` which extracts the year, day and ice concentration for each lake and stores them in a structured format. After, the `find_max_ice()` function goes through the stored dataset and compares the daily ice concentrations to find the highest value for each of the six lakes, excluding the great lakes total. If a new maximum is found, the previous stored value is replaced and if this maximum value repeats on multiple days, all of those dates are stored. As the last step, the results, containing the lake name, concentration and corresponding date, are displayed.

Output:

Q3 - Maximum Ice Concentration for Each Lake

Superior:

Maximum ice concentration: 11.95%

Occurred on 1 day(s):

2024 day 050

Michigan:

Maximum ice concentration: 18.28%
Occurred on 1 day(s):
2024 day 021

Huron:
Maximum ice concentration: 22.65%
Occurred on 1 day(s):
2024 day 022

Erie:
Maximum ice concentration: 35.72%
Occurred on 1 day(s):
2024 day 022

Ontario:
Maximum ice concentration: 5.30%
Occurred on 1 day(s):
2024 day 022

St. Clair:
Maximum ice concentration: 97.15%
Occurred on 2 day(s):
2024 day 022
2024 day 023

Output screenshot:

```
Q3 - Maximum Ice Concentration for Each Lake
Superior:
Maximum ice concentration: 11.95%
Occurred on 1 day(s):
  2024 day 050

Michigan:
Maximum ice concentration: 18.28%
Occurred on 1 day(s):
  2024 day 021

Huron:
Maximum ice concentration: 22.65%
Occurred on 1 day(s):
  2024 day 022

Erie:
Maximum ice concentration: 35.72%
Occurred on 1 day(s):
  2024 day 022

Ontario:
Maximum ice concentration: 5.30%
Occurred on 1 day(s):
  2024 day 022

St. Clair:
Maximum ice concentration: 97.15%
Occurred on 2 day(s):
  2024 day 022
  2024 day 023
```

Question 4:

Output:

Question 4:

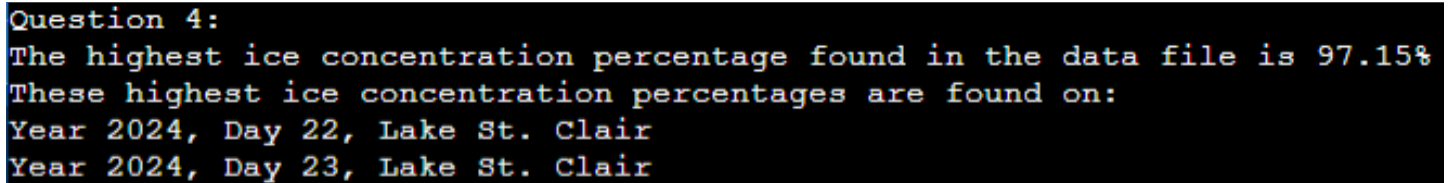
The highest ice concentration percentage found in the data file is 97.15%

These highest ice concentration percentages are found on:

Year 2024, Day 22, Lake St. Clair

Year 2024, Day 23, Lake St. Clair

Output Screenshot:

A screenshot of a terminal window with a black background and white text. The text displays the output for Question 4, showing the highest ice concentration percentage and the specific dates and lake names where it was found.

```
Question 4:  
The highest ice concentration percentage found in the data file is 97.15%  
These highest ice concentration percentages are found on:  
Year 2024, Day 22, Lake St. Clair  
Year 2024, Day 23, Lake St. Clair
```

Procedure:

To answer this question, a significant amount of techniques were utilized from Lesson 08: Strings and Plotting. Firstly, this question involved using a C-Program to create and read from a file. Furthermore, the 'fgets' function was used to ignore the first 5 lines in the text file. A mixture of the while loop and 'fgets' functions were also used to read the file till the end of the file had been reached, furthermore, the contents within the file were stored into a buffer 'line,' the text file was read until the 'fgets' function returned a NULL value. Another function utilized was the 'sscanf' function; values extracted from the text file were stored in the addresses of specific variables. The use of 'if statement' allowed us to check if the number of variables extracted was appropriate, and if they were, then the program was continued. Furthermore, multiple arrays were used; a character type array was used to store the lake's names, and a type float array was used to store the concentration values of the lakes. For loops also played a significant role in completing this question; with the help of the for loop, each concentration in the text file was able to be compared to determine the highest concentration contained in the text file. Lastly, the 'strcpy' function was used to count the length of the buffer line to see if there were appropriate characters in it, ignoring the redundant values.

Question 5:

Procedure:

This question asked for daily ice concentration graphs for each of the six great lakes. The program reads the ice concentration data from the file and separates the data for each great lake into separate program files, each containing a column with day numbers and another column with ice concentrations in percent. It begins by opening the original data file and skipping the header lines to ensure the file is read correctly and there are no issues with displaying the data. Then, the program opens six separate output files. It goes through each line of the original data file and extracts the day and ice concentration for each of the lakes. The values are written into the respective output files. The output files were then uploaded to GNUPlot and a script was written for each of the lakes and a graph was generated for each of the lakes.

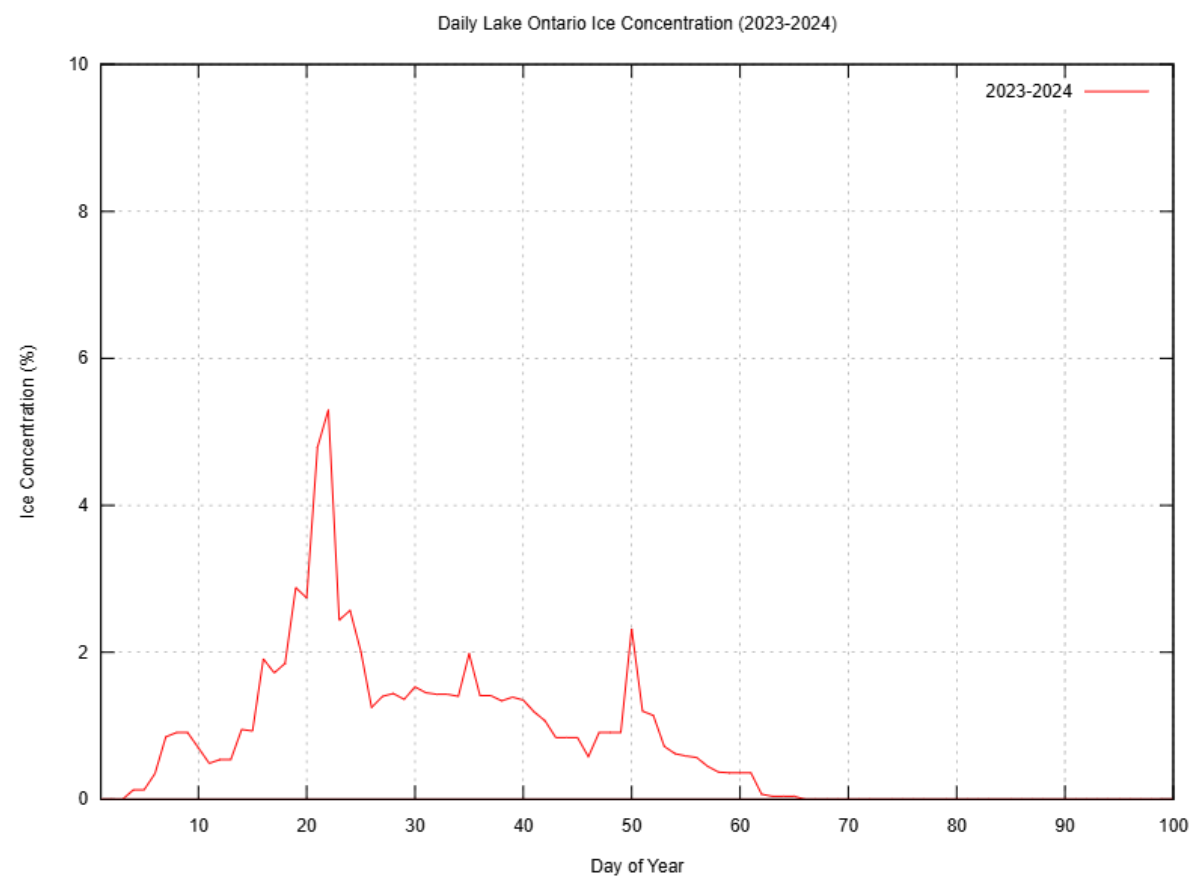
Output:

Daily data files generated for each lake from 2023-2024 data files.

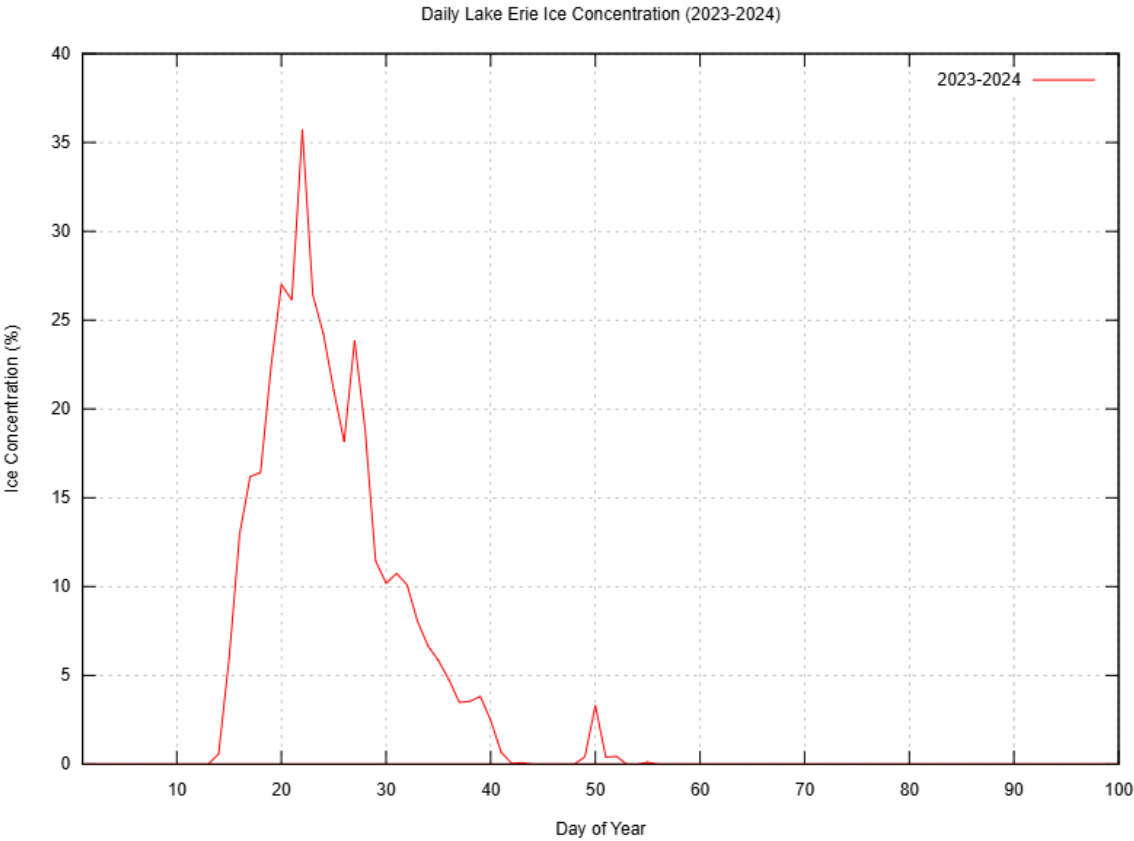
Output Screenshot:

Question 5:
Daily data files generated for each lake from 2023-2024 data files.

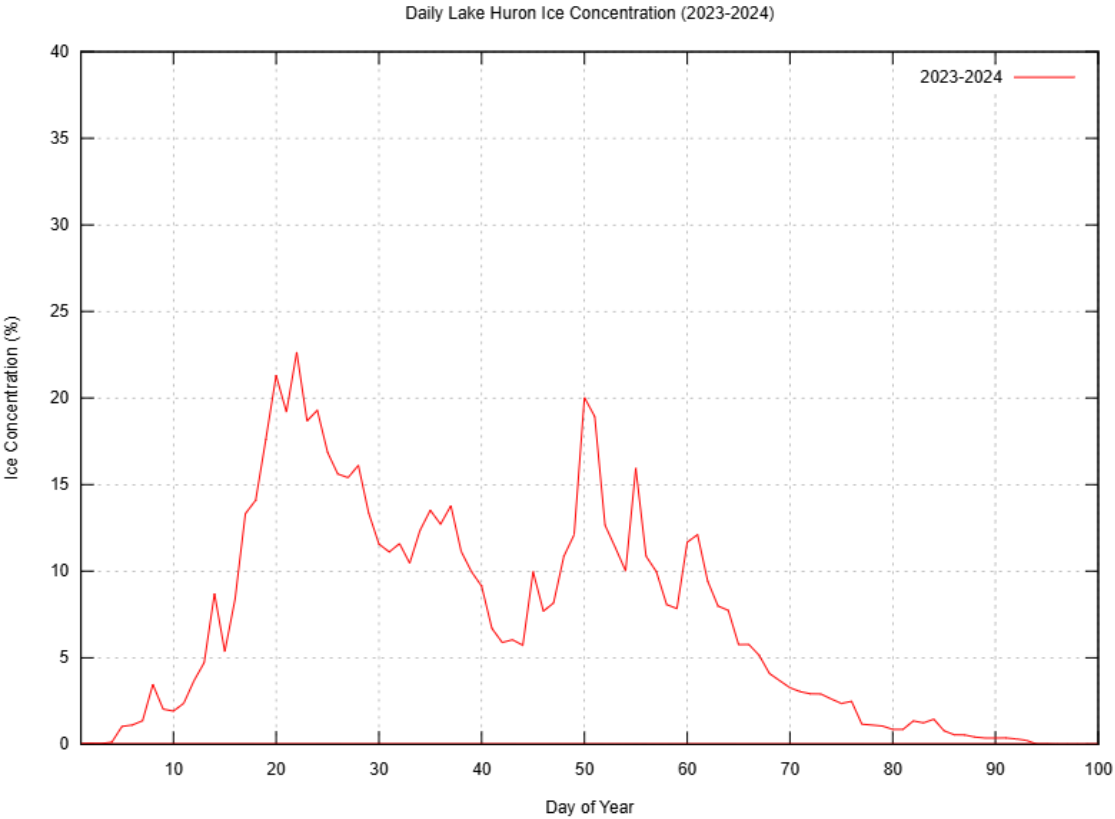
Graph 1: Lake Ontario



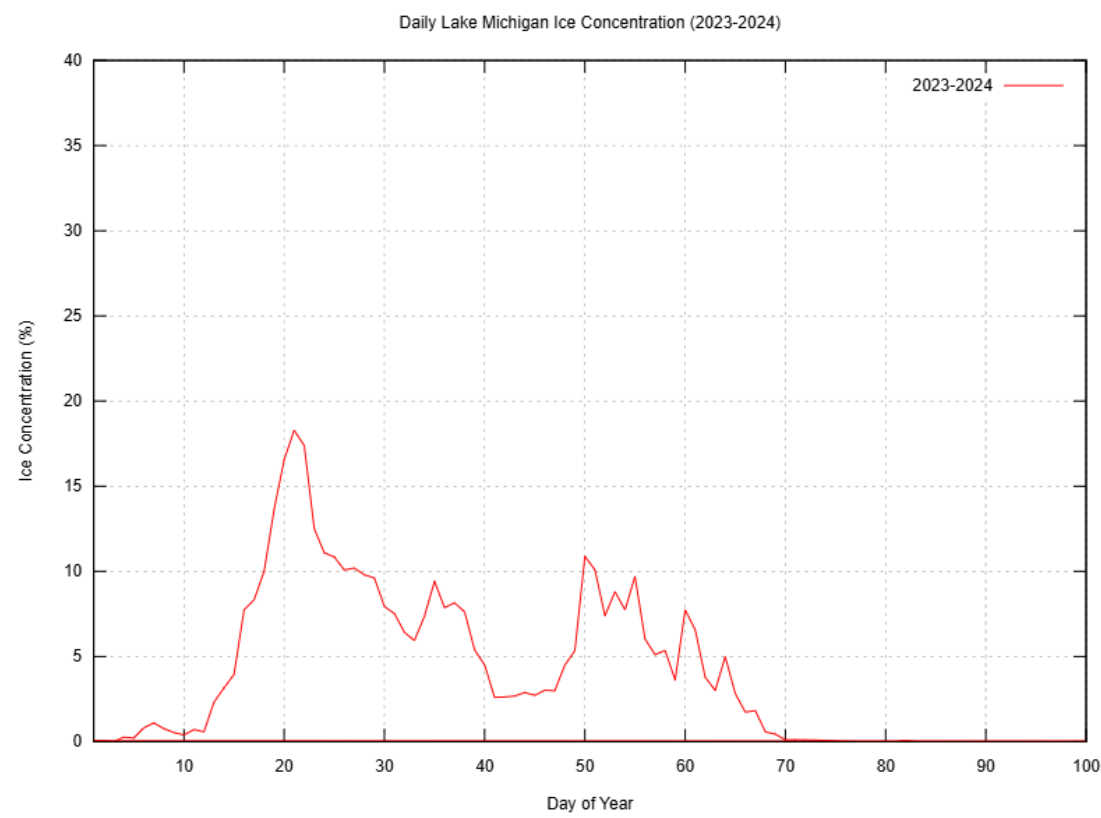
Graph 2: Lake Erie



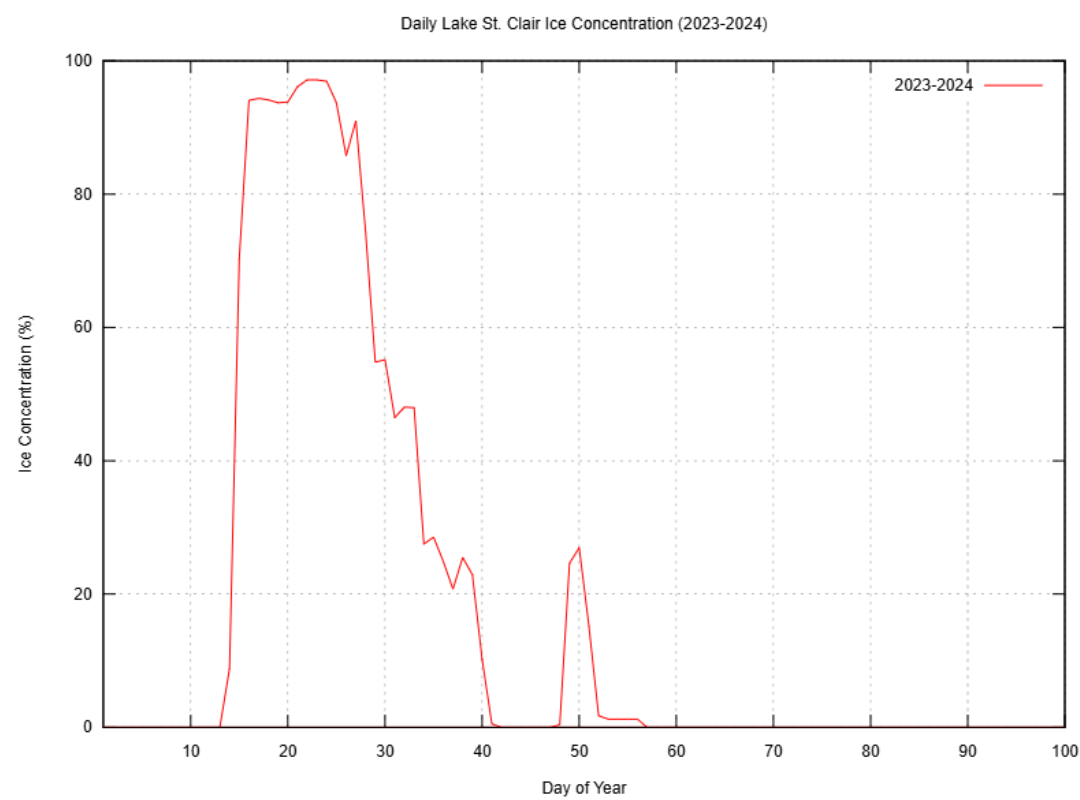
Graph 3: Lake Huron



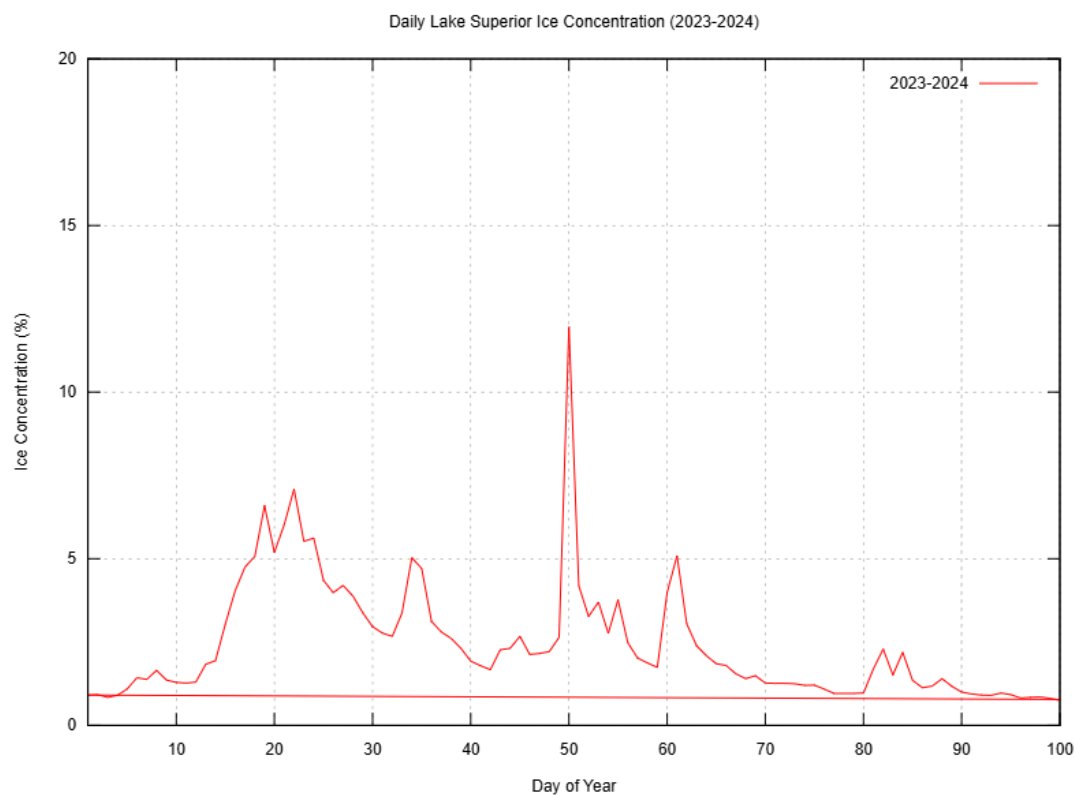
Graph 4: Lake Michigan



Graph 5: Lake St. Clair



Graph 6: Lake Superior



Question 6:

Output:

Question 6:
Data needed to make a single graph of the ice concentration percentages has been added to the 'graphfile.txt' file.

Output Screenshot:

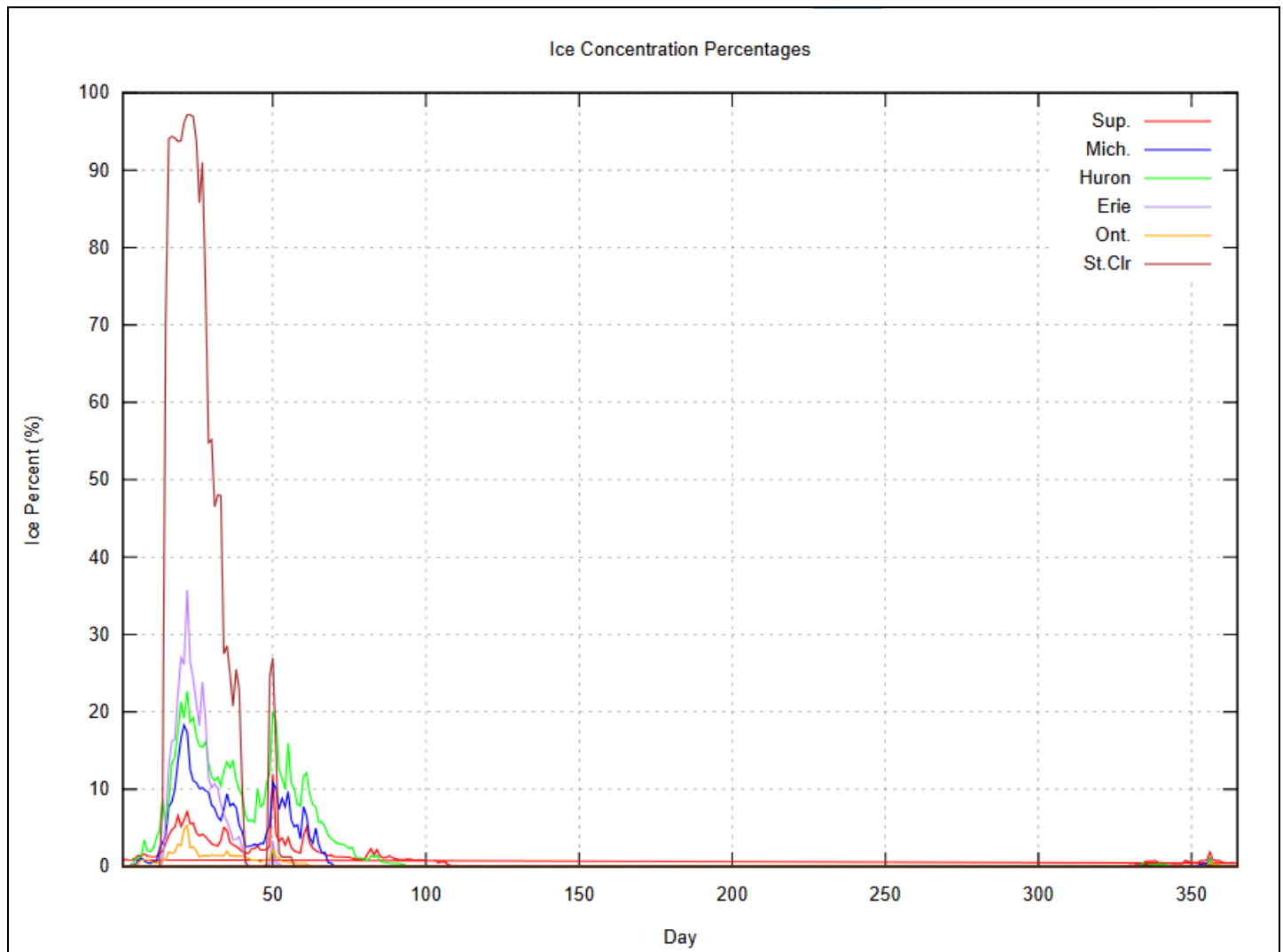
```
Question 6:  
Data needed to make a single graph of the ice concentration  
percentages has been added to the 'graphfile.txt' file.
```

Table 1: Sample Data Stored in ‘graphfile.txt’

Day	Sup.	Mich	Huron	Erie	Ont.	St.Clr
329	0.06	0.00	0.03	0.00	0.00	0.00
330	0.19	0.00	0.03	0.00	0.00	0.00
331	0.11	0.00	0.02	0.00	0.00	0.00
332	0.31	0.00	0.02	0.00	0.00	0.00

...	...	...	...	...	...	...
-----	-----	-----	-----	-----	-----	-----

Graph 1:



Procedure:

To complete this question, arrays were utilized to store the names of the lakes, their concentrations, etc. Furthermore, the 'typedef struct' function was also used; this structure was used to store the ice concentrations for the lakes on given days. Additionally, the technique of declaring a file in reading mode was also used in order to read the data in the text file, so it could be rewritten to something that would allow a graph of days versus concentration percentages to be created. Moreover, 'fgets' function was used to skip the header lines contained in the text file, so the values of the text file that are needed to create the graph could be obtained. In this question, we further dealt with declaring a file in writing mode, and to write in the file 2D arrays were used since they allowed data to be written in the format which it was needed in matrix format, so the graphing could be accomplished in gnuPlot. Besides the C-programming, GNUPlot operations were also used, the knowledge gained from lab 07 was helpful in order to complete the GNUPlot operations in which the title of the graph and the x and y-range of the graph were set. Furthermore, data obtained from the C-programming was utilized to create the graph needed.

Question 7:

Output:

Question 7:

Lake | 2021 Ave % | 2022 Ave % (21-22) | 2022 Ave % (22-23) | 2023 Ave %

Superior	0.88	21.59	1.08	4.71
Michigan	0.63	12.22	1.17	5.02
Huron	0.70	23.90	1.44	8.66
Erie	0.00	24.96	3.05	2.49
Ontario	0.10	5.45	0.38	1.49
St. Clair	0.12	42.35	16.84	9.33
GL Total	0.64	19.24	1.38	5.29

- Notable Differences Observed:

1. There is a dramatic increase in the percentage of ice concentration between the years 2021 and 2022. For instance, the percentage of ice concentration increases from 0.88% to 21.59% in the Superior lake.
2. Furthermore, another notable difference observed is that the percentage of ice concentration significantly decreases during the time period of 2022 to 2022-2023. For lake Superior it goes from 21.59% to 1.08% and 4.71%.

- These differences may be caused due to:

1. Weather Patterns
 2. Climate Change
 3. Regional Variations
- and etc.

Output Screenshot:

```
Question 7:
Lake      | 2021 Ave % | 2022 Ave % (21-22) | 2022 Ave % (22-23) | 2023 Ave %
-----
Superior  | 0.88      | 21.59              | 1.08                | 4.71
Michigan  | 0.63      | 12.22              | 1.17                | 5.02
Huron     | 0.70      | 23.90              | 1.44                | 8.66
Erie      | 0.00      | 24.96              | 3.05                | 2.49
Ontario   | 0.10      | 5.45               | 0.38                | 1.49
St. Clair | 0.12      | 42.35              | 16.84               | 9.33
GL Total  | 0.64      | 19.24              | 1.38                | 5.29
-----
- Notable Differences Observed:
1. There is a dramatic increase in the percentage of ice concentration between the years 2021 and 2022. For instance, the percentage of
ice concentration increases from 0.88% to 21.59% in the Superior lake.
2. Furthermore, another notable difference observed is that the percentage of ice concentration significantly decreases during the time
period of 2022 to 2022-2023. For lake Superior it goes from 21.59% to 1.08% and 4.71%.
- These differences may be caused due to:
1. Weather Patterns
2. Climate Change
3. Regional Variations
and etc.
```

Procedure:

To answer this question, there were many C operations utilized. In this question, I chose to use pointers and this is because the utilization of pointers made this program more efficient, since it has fewer lines of code. Moreover, arrays were also used to complete this question. More specifically, arrays were used to store the total sum of each lake's ice concentration, and the average percent concentration of each lake in each year. These values, which include the total sum of ice concentration and average ice concentration for each lake per year

were calculated with the help of 'if statements', 'for-loops', and counter variables. For instance, to calculate the average ice concentration for each lake per year, the total sum of the concentrations per year was calculated first with the help of arrays and if statements; the if statements made sure what year was being looked at, and the array stored the sum of ice concentrations for each lake. The counter represented the number of days, and both values of the total sum and the counter allowed us to calculate the average percentage ice concentration for each lake per year → $\text{average} = (\text{sum of ice concentrations in one year}) / \text{counter}$ (which represented days).

Question 8:

Procedure:

The main task of this question was to process seasonal ice concentration data from three different data sets from 2021 to 2022, 2022 to 2023, and 2023 to 2024, and produce daily data files for each of six Great Lakes over a 365 day period from late fall of one year to early spring of another year. We created a function named (extract_daily_data) which reads input files, skips header lines, and parses each line for year, day and concentrations using sscanf. The days were also adjusted from late fall to early spring for smoother analysis and saving data files for each lake separately. Then these values are stored in a 3D array named (data[season_number][adjusted_day][lake]) with season numbered 0,1, and 2. In the main program, the function was called for each season's file, populating the arrays. Therefore, six output files were created, each with a header and 365 lines listing the day and concentration for all the lakes. The output files were named after the name of the lakes with extension of dat. After executing the C Program, we downloaded the files and added them in GNUplot and plotted the data by writing an organized script. After running the script in GNUplot, the graphs were generated which enabled us to show a detailed visualization of seasonal ice trends across the Great Lakes over three years.

Output:

Question 8:

Daily data files generated for each lake.

```
Question 8:
Daily data files generated for each lake.

...Program finished with exit code 0
Press ENTER to exit console. □
```

Table 1: Sample Data Stored in ‘Superior.dat’

Day	2021-2022	2022-2023	2023-2024
1	0.01	0.29	0.19
2	0.17	0.20	0.11
3	0.12	0.27	0.31
4	0.13	0.22	0.41
...	...	...	...

Graph 1:

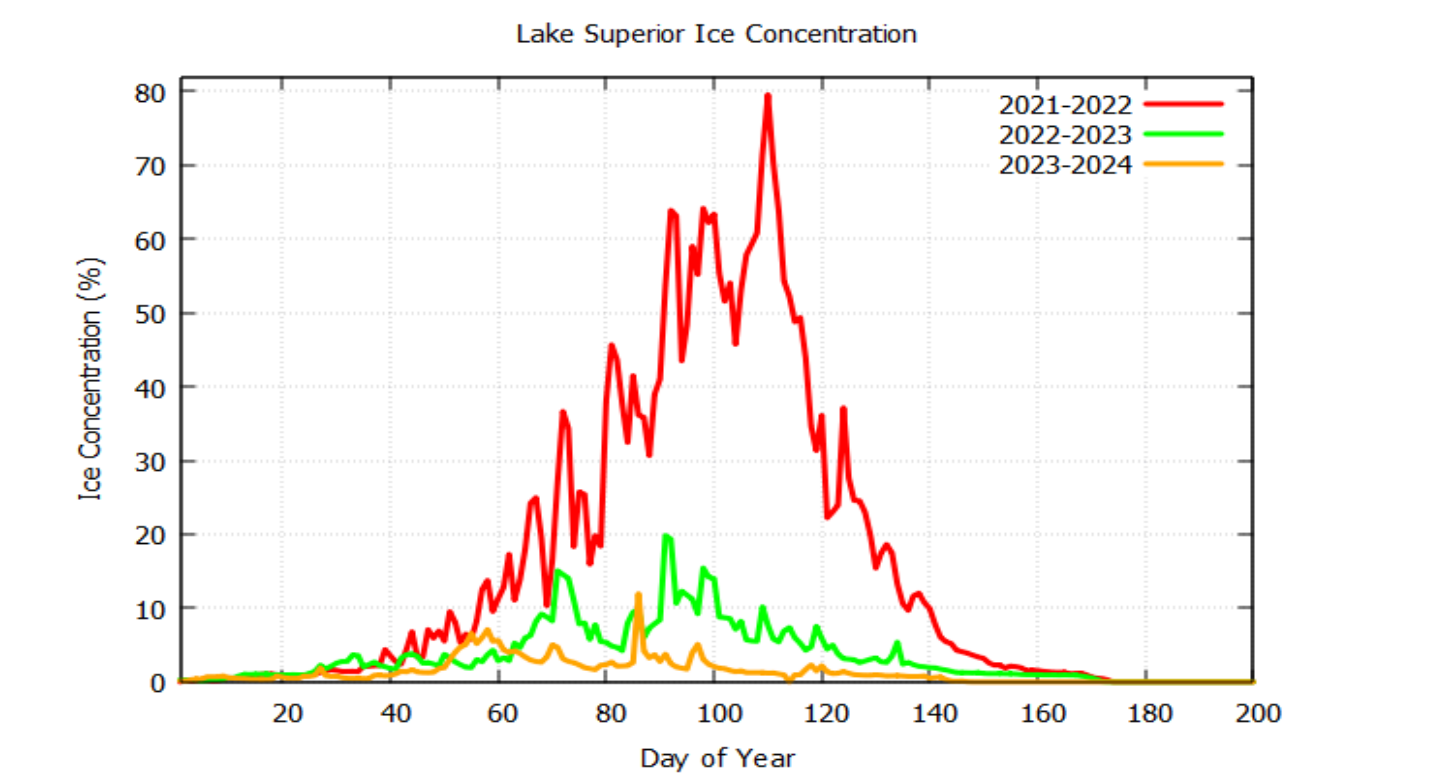


Table 2: Sample Data Stored in ‘Ontario.dat’

Day	2021-2022	2022-2023	2023-2024
28	0.03	0.66	0.28
29	0.65	0.92	0.25
30	0.65	1.78	0.25
31	0.37	1.66	0.04
...	...	...	...

Graph 2:

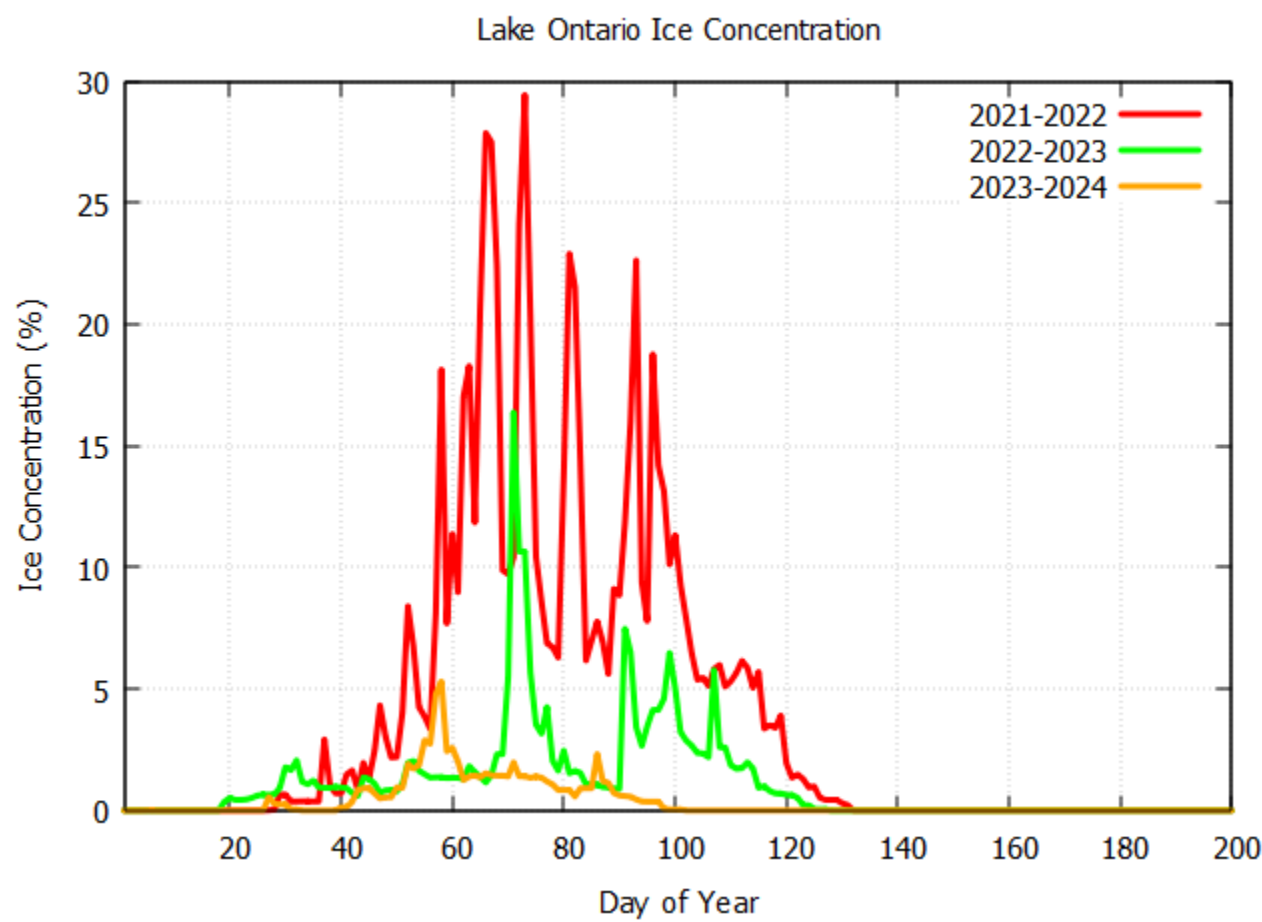


Table 3: Sample Data Stored in ‘Huron.dat’

Day	2021-2022	2022-2023	2023-2024
1	0.00	0.17	0.03
2	0.00	0.03	0.02
3	0.00	0.03	0.02
4	0.02	0.03	0.03
...	...	...	...

Graph 3:

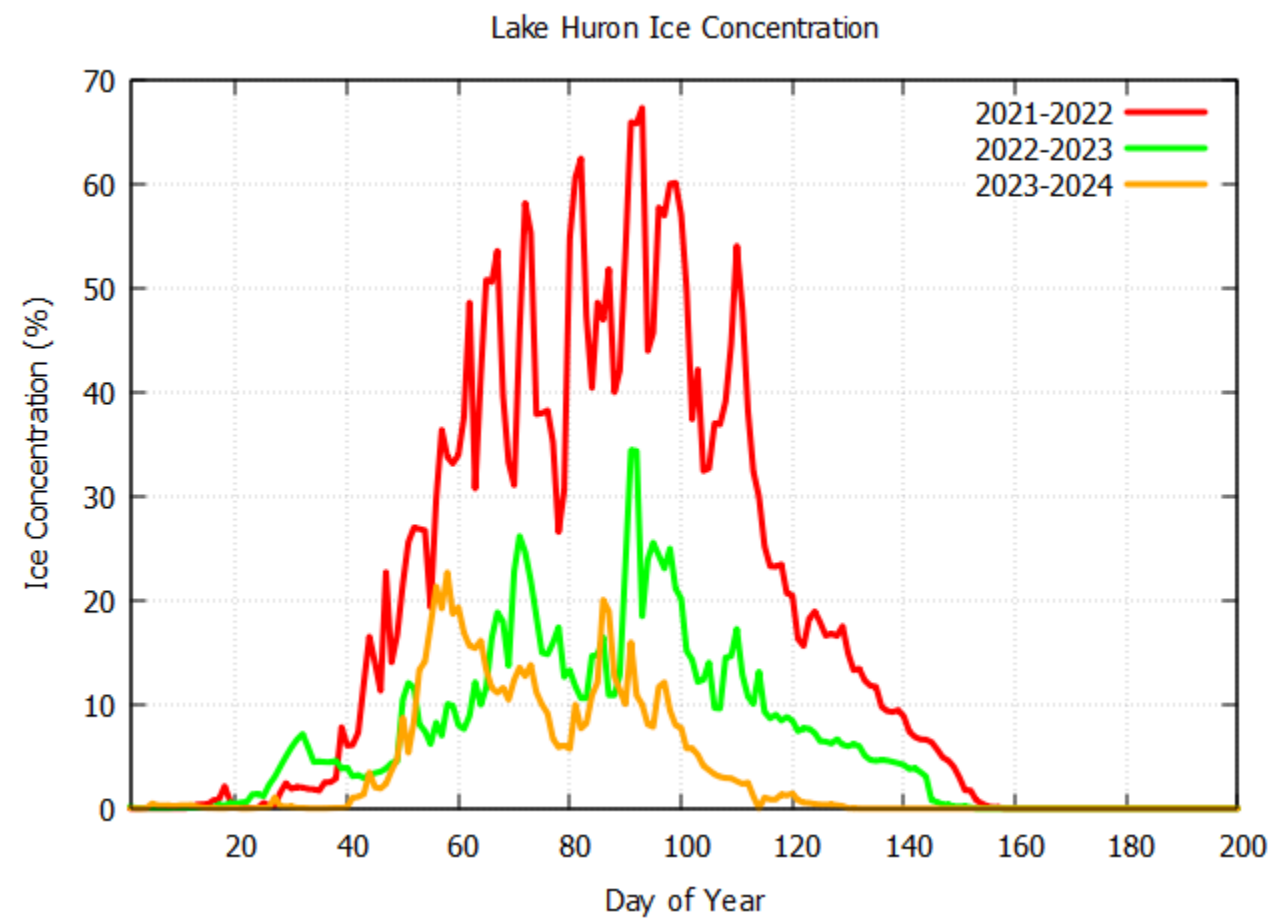


Table 4: Sample Data Stored in ‘Michigan.dat’

Day	2021-2022	2022-2023	2023-2024
10	0.19	0.25	0.13
11	0.42	0.25	0.09
12	0.03	0.18	0.04
13	0.62	0.22	0.04
...	...	...	...

Graph 4:

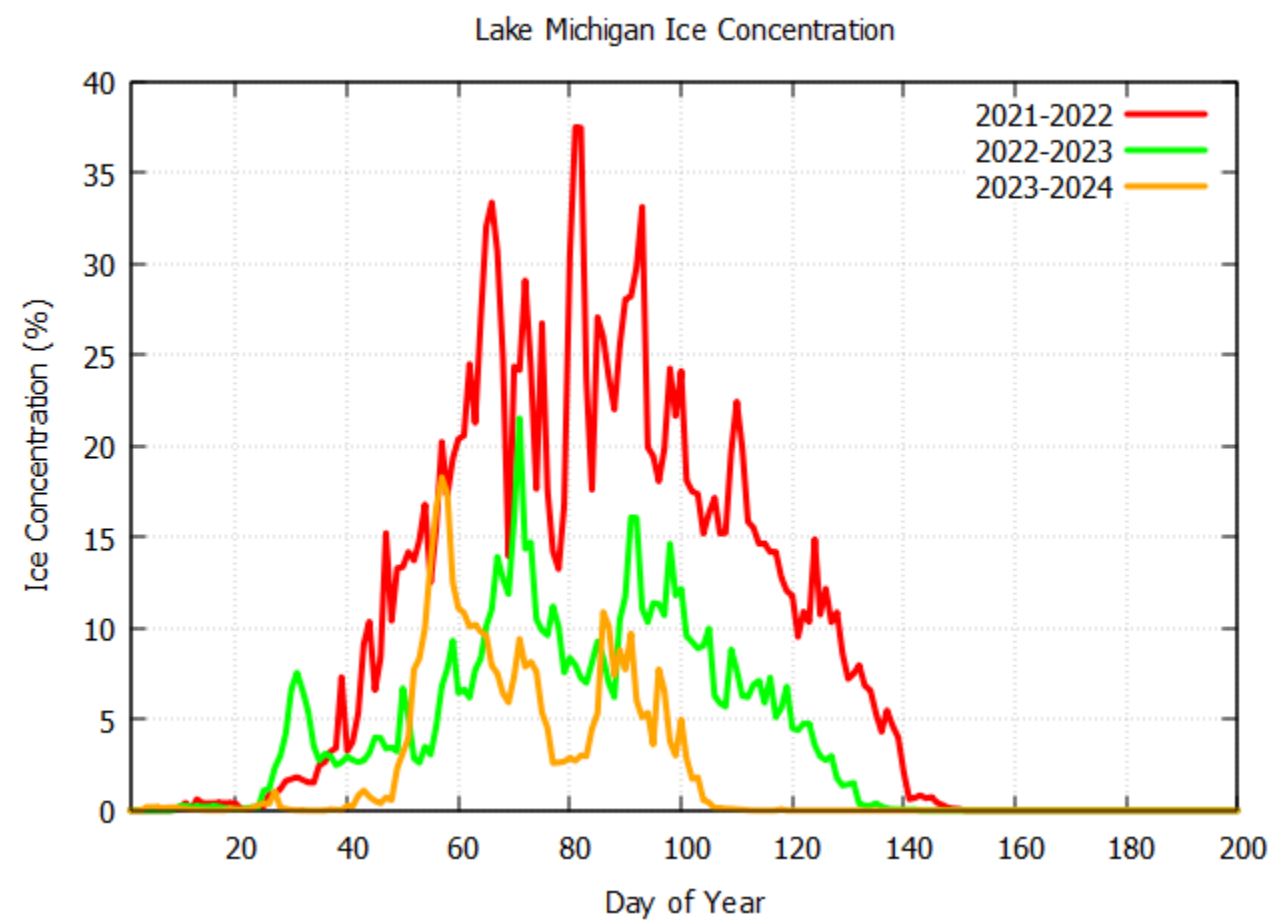


Table 5: Sample Data Stored in ‘Erie.dat’

Day	2021-2022	2022-2023	2023-2024
41	0.13	5.18	0.00
42	3.36	4.11	0.00
43	6.64	4.43	0.00
44	6.68	4.14	0.00
...	...	...	...

Graph 5:

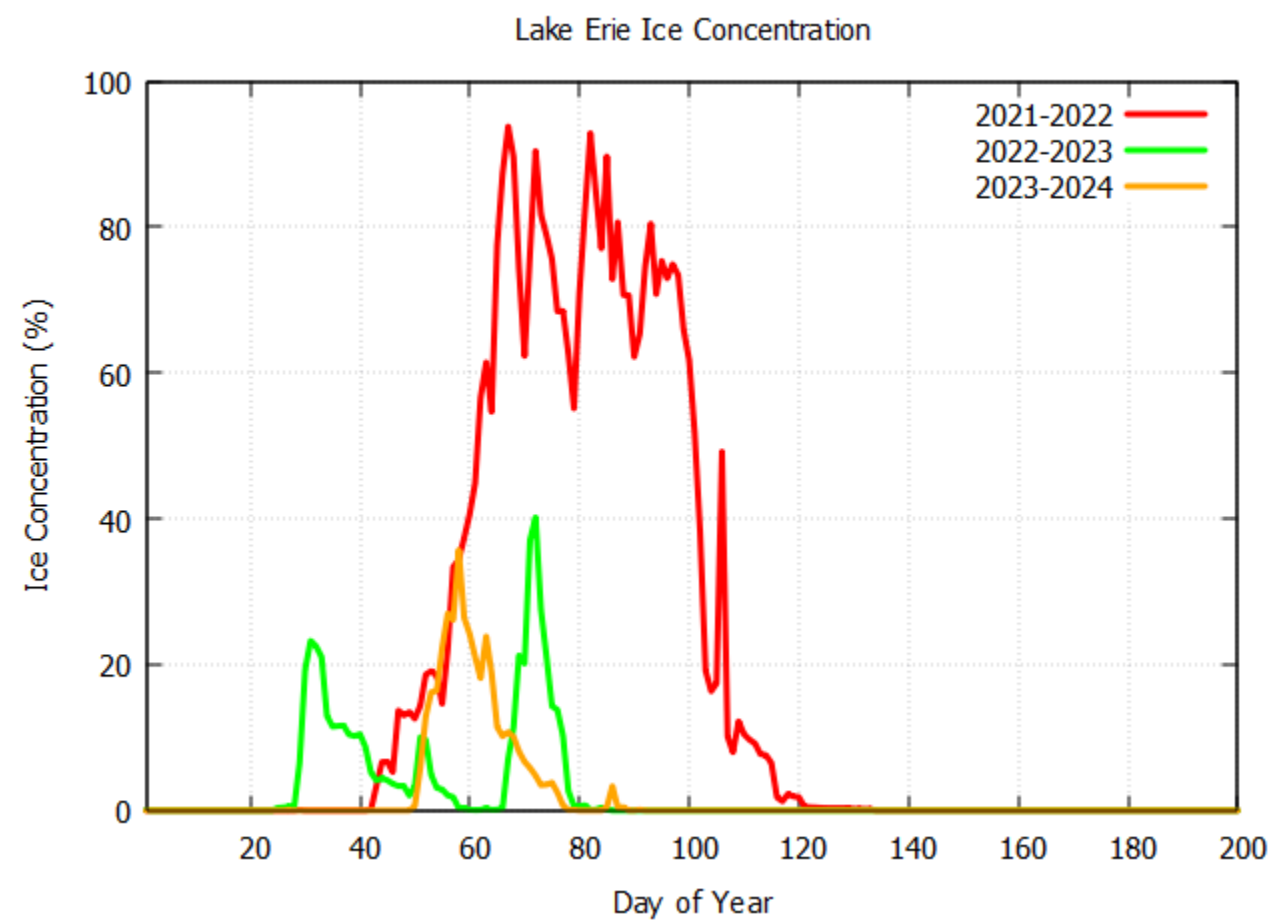
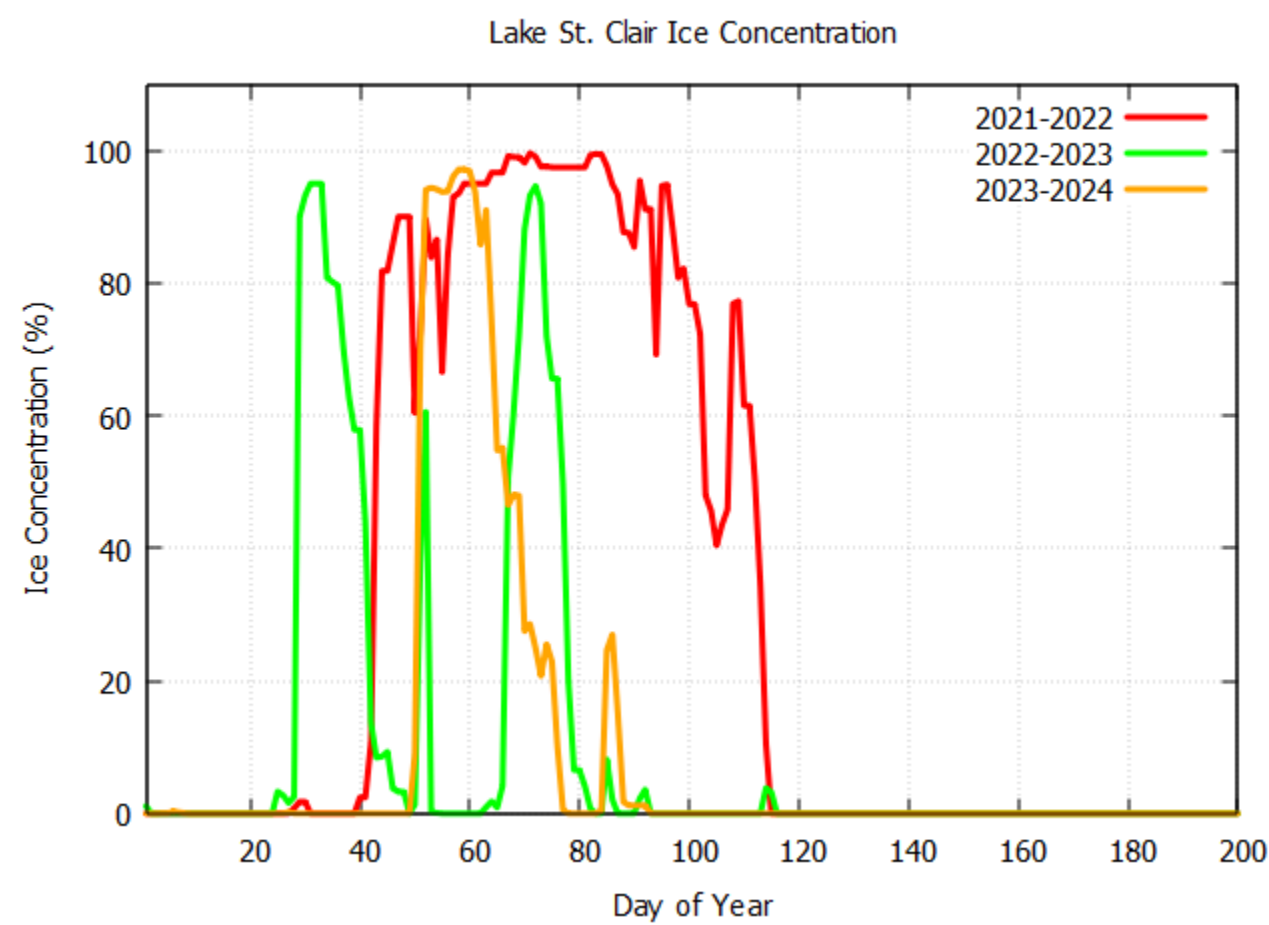


Table 6: Sample Data Stored in ‘stclair.dat’

Day	2021-2022	2022-2023	2023-2024
28	0.80	2.54	0.00
29	1.74	90.00	0.00
30	1.74	93.20	0.00
31	0.00	95.00	0.00
...	...	...	...

Graph 6:



Conclusion: Working on this project was a valuable experience for all members of this group. It provided valuable experience in handling real-world data using C programming and GNUPlot. It allowed us to practice effective teamwork and communication, especially when issues arose and we were required to work closely together to solve the problem. Challenges rose frequently when plotting the GNUPlot graphs as we all had little experience working with the graphing software. We spent some time going over each other's work to ensure everything was done correctly and helped each other in solving problems, such as filling in missing parts of the GNUPlot scripts that prevented the graph from being correctly generated. If we could go back and redo this project we would do some things differently. For example, we would have started looking for a fourth member sooner, as having an additional member would have lessened the workload for each member and would have allowed more minds to be put into answering the questions. We would also have certainly started sooner, as we did leave some parts of the project to the last minute and had to make some sacrifices to ensure it got submitted in time.